

Death and Taxes: Inheritance Tax Planning and Unexpected Mortality

Yonatan Berman[†] Jeanne Bomare[‡] Ignacio Orueta[‡]

[†]King's College London, London School of Economics and IZA

[‡]London School of Economics

NICEP Conference, 12 May 2026

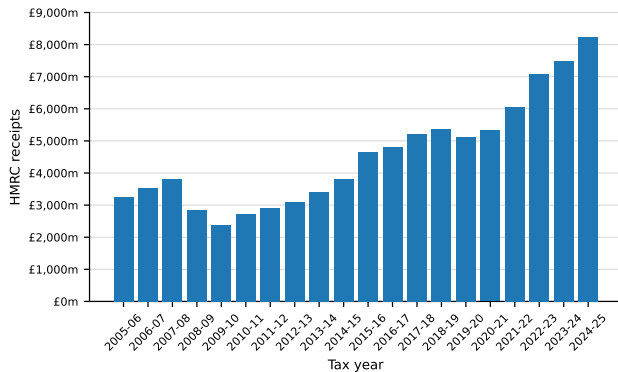
HMRC Disclaimer

This work contains statistical data from HMRC which is Crown Copyright. The research datasets used may not exactly reproduce HMRC aggregates. The use of HMRC statistical data in this work does not imply the endorsement of HMRC in relation to the interpretation or analysis of the information.

Motivation

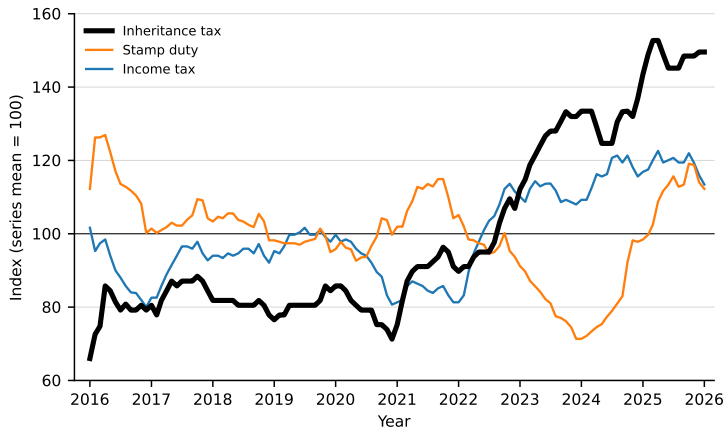
► Inheritance Tax (IHT) is increasingly salient in the UK

IHT receipts, tax year 2005/2006 to tax year 2024/2025



Motivation

► Inheritance Tax (IHT) is increasingly salient in the UK



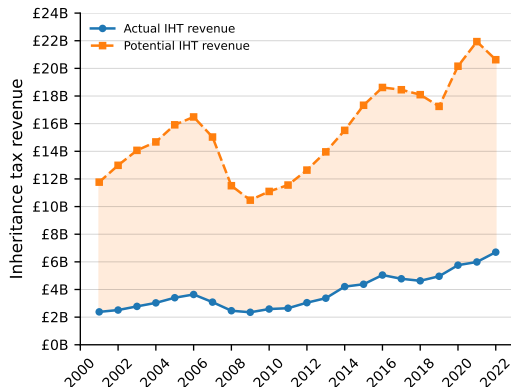
Source: Authors' calculations using Google Trends data

Motivation

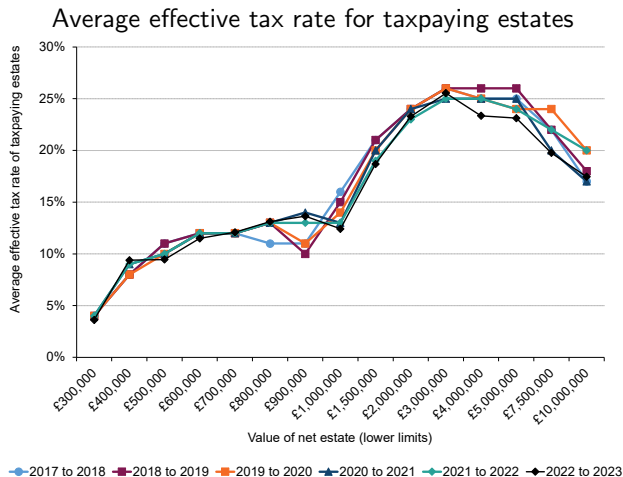
- **Inheritance Tax (IHT) is highly concentrated:** only 4% of estates pay IHT, but potentially high tax bills at stake → vulnerable to planning.

Motivation

- **Inheritance Tax (IHT) is highly concentrated:** only 4% of estates pay IHT, but potentially high tax bills at stake → vulnerable to planning.



Motivation



This paper

- ↪ **Estimates the extent of inheritance tax planning in the United Kingdom**
- ↪ **Detects the key planning mechanisms**
- ↪ **Analyzes the impact of planning on inequality (not today)**

This paper

- ↪ **Estimates the extent of inheritance tax planning in the United Kingdom**
- ↪ **Detects the key planning mechanisms**
- ↪ **Analyzes the impact of planning on inequality (not today)**

Data

- ▶ Administrative inheritance tax data
- ▶ Administrative data on mortality

This paper

- ↪ **Estimates the extent of inheritance tax planning in the United Kingdom**
- ↪ **Detects the key planning mechanisms**
- ↪ **Analyzes the impact of planning on inequality (not today)**

Data

- ▶ Administrative inheritance tax data
- ▶ Administrative data on mortality

Methodology

- ▶ Use the first wave of COVID-19 deaths as a quasi-experimental shock in sudden deaths
- ▶ Sudden deaths reduce opportunities for tax planning
- ▶ Leverage age, gender and regional variation in excess mortality to measure the extent of tax planning in the UK in a quasi difference-in-differences setting

Literature

- ❶ **Scant evidence on the extent of inheritance tax planning** ([Wolff, 1997](#); [Poterba, 2000](#); [Kopczuk, 2013](#))
 - ▶ Quasi-experimental setting exploiting a large exogenous shock in mortality
 - ▶ Focus on the UK
 - ▶ Implications for designing and collecting a wealth tax
- ❷ **Inheritance tax planning strategies** ([Moretti and Wilson, 2023](#); [Micó, 2024](#); [Sturrock et al., 2025](#))
 - ▶ Use of IHT reliefs in the UK ([Advani and Sturrock, 2023](#))
 - ▶ Importance of other channels for planning in the UK (e.g., gifts)
- ❸ **Estimation of wealth inequality using inheritance tax data** ([Alvaredo et al., 2018, 2025](#); [Acciari et al., 2024](#))

Key results

- ▶ Inheritance tax planning reduces government revenue by £3–£4.5 billion each year (55%–110% of receipts)
- ▶ Reducing the value of assets held at death (e.g., through inter-vivos gifts) has a greater impact on reducing IHT liability than making strategic use of reliefs

Institutional setting

Threshold and Rate:

- ▶ IHT is levied on the total value of a decedent's estate at death + inter-vivos gifts
- ▶ *Threshold*: £325,000
- ▶ *Standard rate*: 40% on value above the threshold

Institutional setting

Threshold and Rate:

- ▶ IHT is levied on the total value of a decedent's estate at death + inter-vivos gifts
- ▶ *Threshold*: £325,000
- ▶ *Standard rate*: 40% on value above the threshold

Main Reliefs and Allowances:

- ▶ *Spouse exemption*: transfers between spouses and civil partners are exempt
- ▶ *Residence nil-rate band*: additional allowance when passing the main residence to direct descendants
- ▶ *Business and agricultural reliefs*: reduce the taxable value of qualifying assets
- ▶ *Seven-year rule*

Institutional setting

Threshold and Rate:

- ▶ IHT is levied on the total value of a decedent's estate at death + inter-vivos gifts
- ▶ *Threshold*: £325,000
- ▶ *Standard rate*: 40% on value above the threshold

Main Reliefs and Allowances:

- ▶ *Spouse exemption*: transfers between spouses and civil partners are exempt
- ▶ *Residence nil-rate band*: additional allowance when passing the main residence to direct descendants
- ▶ *Business and agricultural reliefs*: reduce the taxable value of qualifying assets
- ▶ *Seven-year rule*

Revenues:

- ▶ £8 billion in 2024/25 ($\approx 1\%$ of total tax receipts; $< 0.5\%$ of GDP)

Data

① Administrative tax data

- ▶ Inheritance tax returns (IHT400)
- ▶ Universe of taxpaying estates and random sample of non-taxpaying estates (covering in total 40%–50% of annual deaths)
- ▶ Covers the 2011–2022 period
- ▶ Information on decedents:
 - ▶ Demographic characteristics (gender, age, last address, etc.)
 - ▶ Size and composition of the taxable estate at death
 - ▶ IHT reliefs claimed
 - ▶ Total inheritance tax due

Data

① Administrative tax data

- ▶ Inheritance tax returns (IHT400)
- ▶ Universe of taxpaying estates and random sample of non-taxpaying estates (covering in total 40%–50% of annual deaths)
- ▶ Covers the 2011–2022 period
- ▶ Information on decedents:
 - ▶ Demographic characteristics (gender, age, last address, etc.)
 - ▶ Size and composition of the taxable estate at death
 - ▶ IHT reliefs claimed
 - ▶ Total inheritance tax due

② Mortality data

- ▶ Office for National Statistics (ONS) data on excess mortality over time
- ▶ Weekly measure, available for the first COVID wave (April–May 2020)
- ▶ Disaggregated by age group, gender, and location

Methodology I

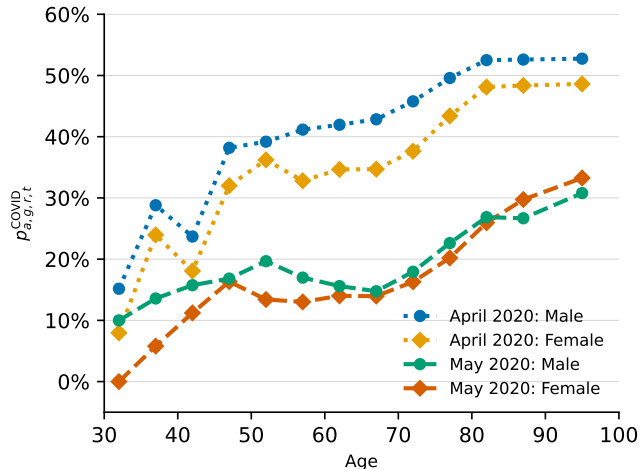
Identify unexpected deaths

- ▶ Use the first wave of COVID-19 deaths in 2020 as an exogenous mortality shock
- ▶ Increases the probability of dying unexpectedly, limiting opportunities for estate planning
- ▶ Compute the conditional probability that an individual who died in April–May 2020 died unexpectedly, for each age group a , gender g , region r , and month t :

$$p_{a,g,r,t}^{\text{COVID}} = \frac{\text{Actual_Deaths}_{a,g,r,t} - \text{Predicted_Deaths}_{a,g,r,t}}{\text{Actual_Deaths}_{a,g,r,t}}$$

- ▶ Set $p_{a,g,r,t}^{\text{COVID}} = 0$ outside the first wave of COVID-19 and for decedents younger than 30 or older than 80 years (in our preferred specification)

Conditional unexpected death probability by age and gender



Source: Authors' calculations using data from the ONS

Methodology II

Quasi difference-in-differences setting

- ▶ Construct a panel by averaging key variables from individual tax returns within each age group (a) and gender (g) for each month (t)
- ▶ Compute effect of unexpected deaths on IHT in 2020:

$$y_{a,g,t} = \alpha + \delta \cdot p_{a,g,t}^{\text{COVID}} + \theta_t + \kappa_{a,g} + \epsilon_{a,g,t}$$

- ▶ $y_{a,g,t}$: total wealth, tax liability, effective IHT rate, or reliefs claimed
- ▶ θ_t : period (month) fixed effects
- ▶ $\kappa_{a,g}$: age-by-gender fixed effects

Methodology II

Quasi difference-in-differences setting

- ▶ Construct a panel by averaging key variables from individual tax returns within each age group (a) and gender (g) for each month (t)
- ▶ Also estimating placebo effects for previous years:

$$y_{a,g,t} = \alpha + \sum_{y=2016}^{2020} \delta_y \cdot p_{a,g,t}^{\text{COVID}} + \theta_t + \kappa_{a,g} + \epsilon_{a,g,t}$$

- ▶ $y_{a,g,t}$: total wealth, tax liability, effective IHT rate, or reliefs claimed
- ▶ θ_t : period (month) fixed effects
- ▶ $\kappa_{a,g}$: age-by-gender fixed effects

Methodology II

Quasi difference-in-differences setting

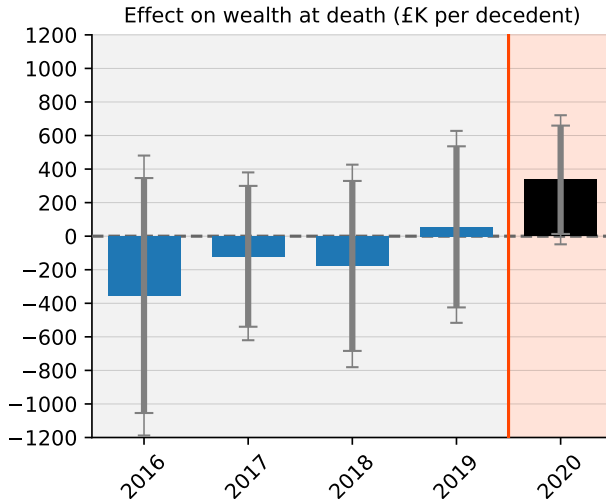
- ▶ Construct a panel by averaging key variables from individual tax returns within each age group (a) and gender (g) for each month (t)
- ▶ Also estimating placebo effects for previous years:

$$y_{a,g,t} = \alpha + \sum_{y=2016}^{2020} \delta_y \cdot p_{a,g,t}^{\text{COVID}} + \theta_t + \kappa_{a,g} + \epsilon_{a,g,t}$$

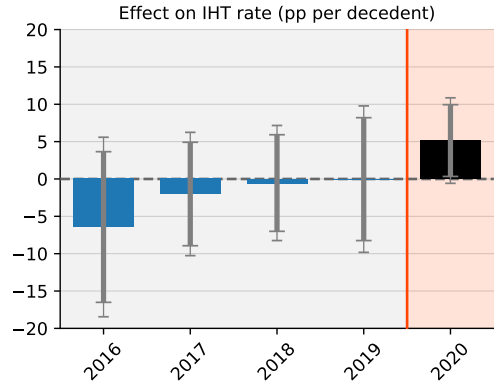
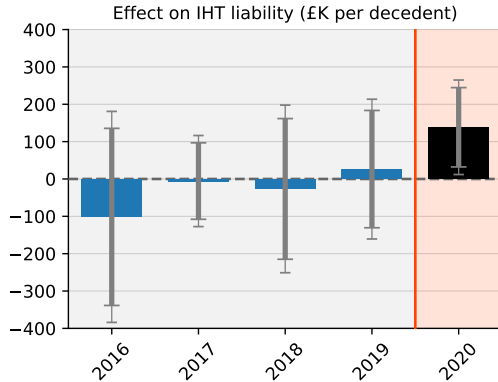
- ▶ $y_{a,g,t}$: total wealth, tax liability, effective IHT rate, or reliefs claimed
 - ▶ θ_t : period (month) fixed effects
 - ▶ $\kappa_{a,g}$: age-by-gender fixed effects
- ▶ Expect the estimated placebo effects $\hat{\delta}_{2016} - \hat{\delta}_{2019}$ to be null

[illegible]

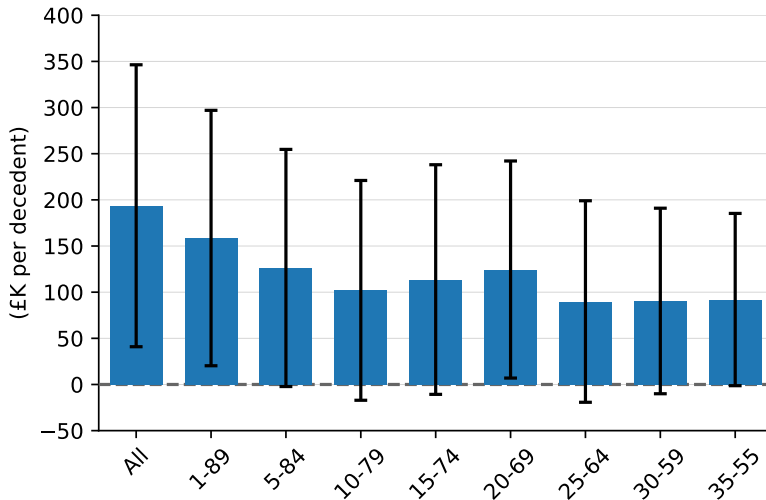
Results: The effect of unexpected death probability on wealth at death



Results: The effect of unexpected death probability on IHT liability and rate



Results: Robustness to treated age group definition



Mechanisms: Estimated effect on wealth components at death

	Total Assets	Savings	Properties	Farm	Business	Other Assets
δ	352119.8* (196164.7)	365042.3*** (133210.3)	-11967.6 (76004.4)	-11841.2 (7347.9)	-29229.7* (16798.8)	38663.9 (40666.5)
Adj. R^2	0.23	0.21	0.25	0.09	0.01	0.03
N	3490	3490	3490	3490	3490	3490

Mechanisms: Estimated effect on claimed reliefs

	Total Reliefs	Business Reliefs	Property Reliefs	Other Reliefs
δ	-37812.9 (29175.4)	-30296.0** (14850.8)	-7,624.5 (6,789.5)	107.6 (21338.3)
Adj. R^2	0.03	0.01	0.25	0.02
N	3490	3490	3490	3490

How much revenue is lost to IHT planning?

A simple model:

$$T_o^i = T_t^i - \beta \cdot \pi \cdot (1 - p^i)$$

- ▶ T_o^i : observed tax due
- ▶ T_t^i : true tax due
- ▶ β : average amount of tax reduced by planning
- ▶ π : probability that an individual i planned their estate
- ▶ p^i : probability that i died unexpectedly

How much revenue is lost to IHT planning?

A simple model:

Rearranging:

$$T_o^i = (T_t^i - \beta \cdot \pi) + \beta \cdot \pi \cdot p^i,$$

which closely resembles our regression model:

$$y_i = \alpha_i + \delta p_i$$

How much revenue is lost to IHT planning?

A simple model:

Rearranging:

$$T_o^i = (T_t^i - \beta \cdot \pi) + \beta \cdot \pi \cdot p^i,$$

which closely resembles our regression model:

$$y_i = \alpha_i + \delta p_i$$

How much revenue is lost to IHT planning?

A simple model:

The total amount of tax lost can be found by aggregating over all individuals:

$$\Delta T = \sum_i (T_t^i - T_o^i) = \sum_i \beta \cdot \pi \cdot (1 - p^i) = \sum_i \beta \cdot \pi = \sum_i \hat{\delta} = \hat{\delta} N,$$

where we assumed that $p^i = 0$, i.e., no unexpected deaths

How much revenue is lost to IHT planning?

A simple model:

The total amount of tax lost can be found by aggregating over all individuals:

$$\Delta T = \sum_i (T_t^i - T_o^i) = \sum_i \beta \cdot \pi \cdot (1 - p^i) = \sum_i \beta \cdot \pi = \sum_i \hat{\delta} = \hat{\delta} N,$$

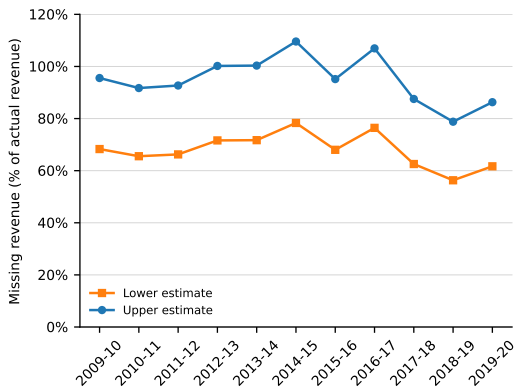
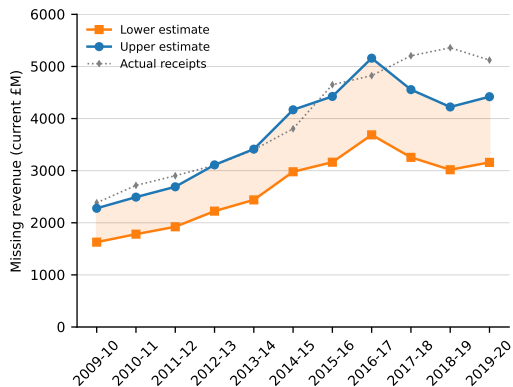
where we assumed that $p^i = 0$, i.e., no unexpected deaths

For 2019:

- ▶ Taxpaying estates: 22,500
- ▶ Estimated $\hat{\delta}$: £140,000

→ Total revenues lost: $\approx$ £3 billion, or 65% of total IHT revenues over the year

Extended to other years



Source: authors' calculations from HMRC data

Conclusion

This paper

- ▶ Uses unexpected deaths as a quasi-experimental shock to planning
- ▶ To identify the extent of IHT planning in the UK

Findings

- ▶ IHT planning is substantial: reduces revenues by £3–£4.5 billion each year (55%–110% of receipts)
- ▶ IHT planning primarily operates through reductions in taxable estate at death (possibly through inter-vivos gifts)
- ▶ rather than via the use of tax-deductible reliefs

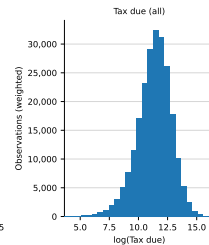
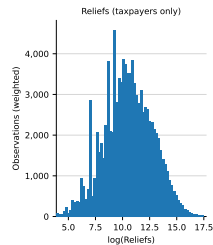
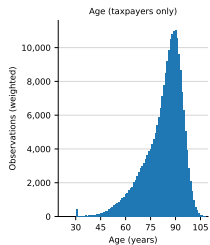
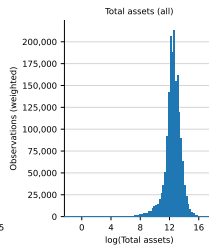
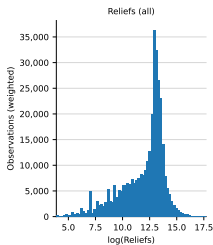
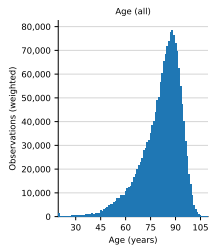
References I

- Acciari, P., Alvaredo, F., and Morelli, S. (2024). The concentration of personal wealth in Italy 1995–2016. *Journal of the European Economic Association*, 22(3):1228–1274.
- Advani, A. and Sturrock, D. (2023). Reforming inheritance tax. Technical report, IFS Report.
- Alvaredo, F., Atkinson, A. B., and Morelli, S. (2018). Top wealth shares in the UK over more than a century. *Journal of Public Economics*, 162:26–47.
- Alvaredo, F., Berman, Y., and Morelli, S. (2025). Evidence from the dead: New estimates of wealth inequality based on the distribution of estates. CEPR Discussion Paper DP19832, CEPR Press.
- Brandily, P., Brébion, C., Briole, S., and Khoury, L. (2021). A poorly understood disease? the impact of COVID-19 on the income gradient in mortality over the course of the pandemic. *European Economic Review*, 140:103923.
- Calderón-Larrañaga, A., Vetrano, D. L., Rizzuto, D., Bellander, T., Fratiglioni, L., and Dekhtyar, S. (2020). High excess mortality during the COVID-19 outbreak in Stockholm region areas with young and socially vulnerable populations. *medRxiv*, pages 2020–07.
- Chen, J. T., Waterman, P. D., Krieger, N., and Krieger, N. (2020). COVID-19 and the unequal surge in mortality rates in Massachusetts. *Population*, 25014(B25014_013E):B25014_001E.
- Chetty, R., Stepner, M., Abraham, S., Lin, S., Scuderi, B., Turner, N., Bergeron, A., and Cutler, D. (2016). The association between income and life expectancy in the United States, 2001–2014. *JAMA*, 315(16):1750–1766.

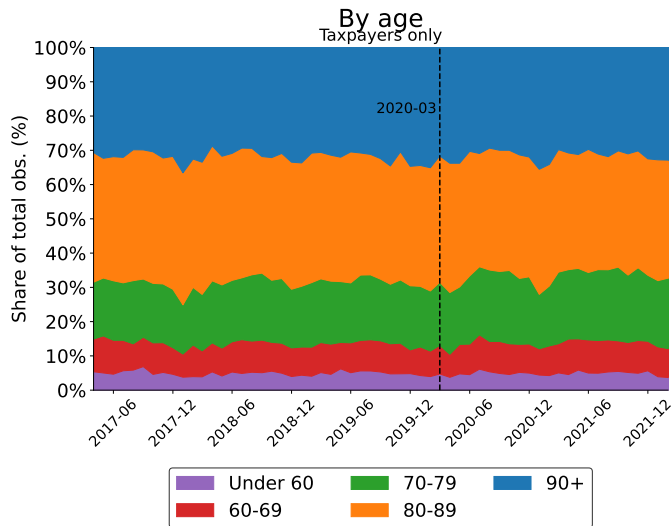
References II

- Decoster, A., Minten, T., and Spinnewijn, J. (2021). The income gradient in mortality during the covid-19 crisis: Evidence from belgium. *Journal of Economic Inequality*, 19(3):551–570.
- Kopczuk, W. (2013). Taxation of intergenerational transfers and wealth. In *Handbook of Public Economics*, volume 5, pages 329–390. Elsevier.
- Micó, I. (2024). Inheritance tax avoidance through the family firm.
- Moretti, E. and Wilson, D. J. (2023). Taxing billionaires: Estate taxes and the geographical location of the ultra-wealthy. *American Economic Journal: Economic Policy*, 15(2):424–466.
- Poterba, J. M. (2000). The estate tax and after-tax investment returns. In Slemrod, J., editor, *Does Atlas Shrug? The Economic Consequences of Taxing the Rich*, pages 329–349. Harvard University Press and Russell Sage Foundation, New York.
- Sturrock, D., Groot, S., and Möhlmann, J. (2025). Wealth, gifts, and estate planning at the end of life. *Journal of Public Economics*, 250:105492.
- Wolff, E. (1997). Discussant comment on douglas holtz-eakin's the uneasy case for abolishing the estate tax. *Tax Law Review*, 51(3):517–521.

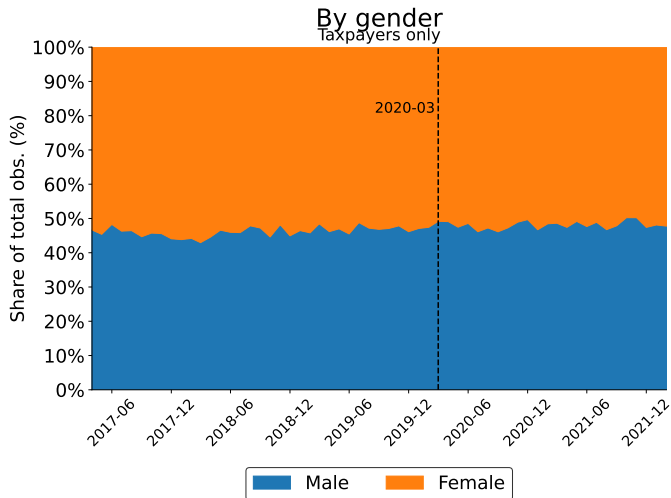
Descriptive statistics [▶ Back](#)



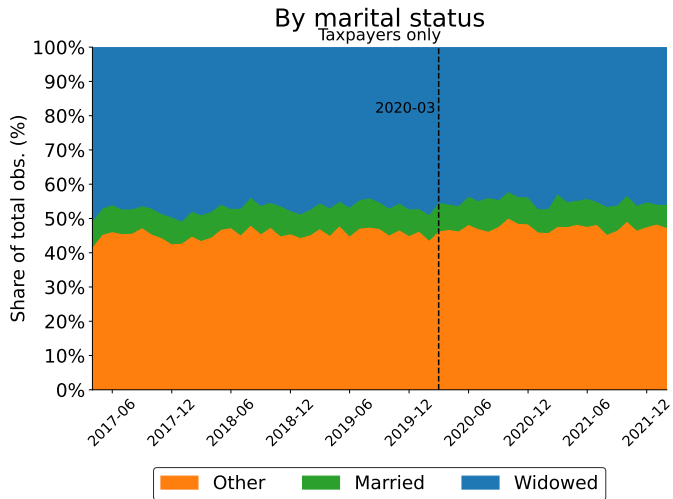
Composition of taxpaying estates remains stable during COVID-19



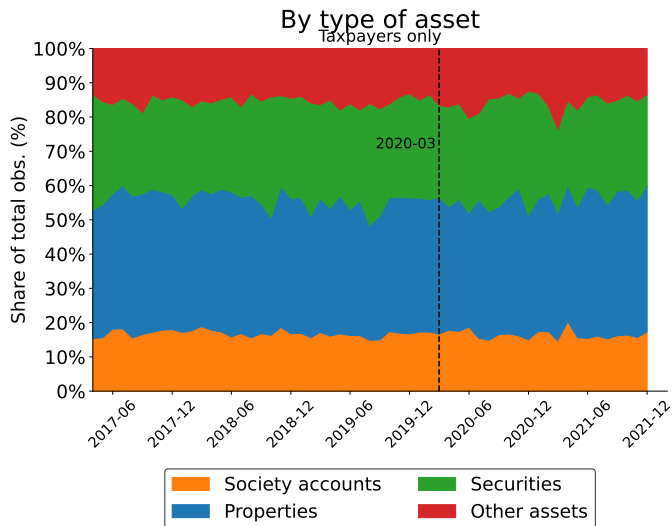
Composition of taxpaying estates remains stable during COVID-19



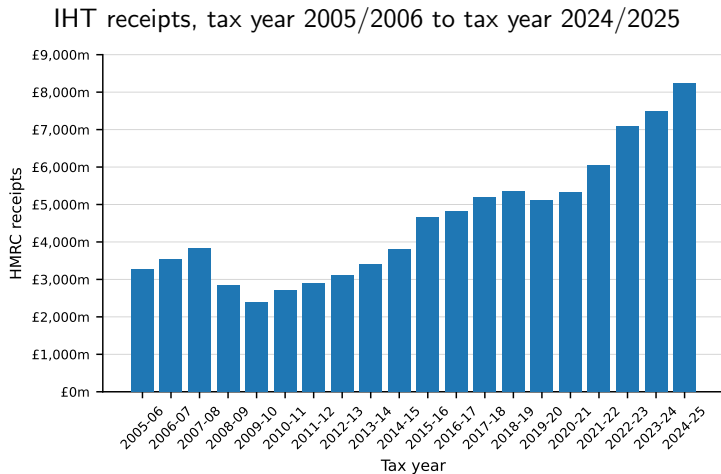
Composition of taxpaying estates remains stable during COVID-19



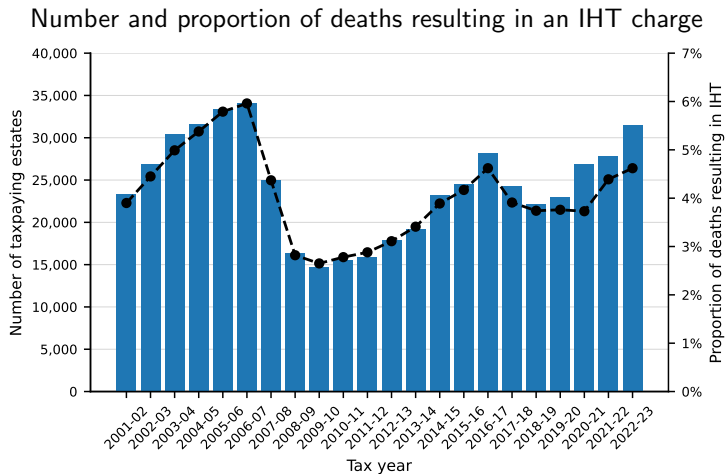
Composition of taxpaying estates remains stable during COVID-19



IHT in the United Kingdom



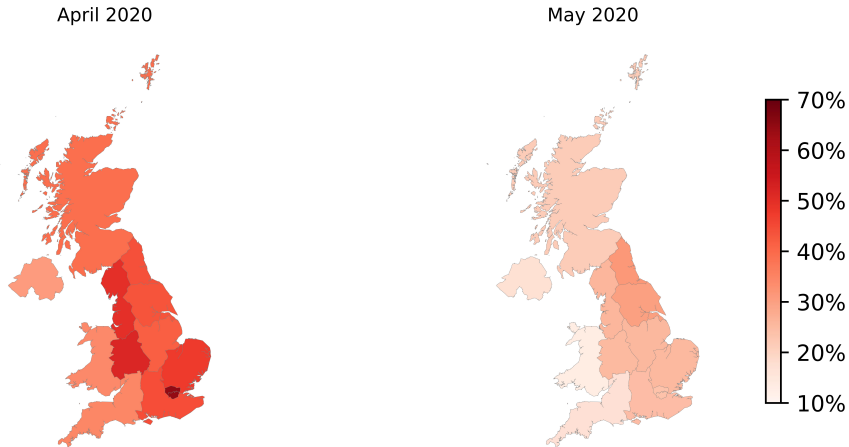
IHT in the United Kingdom



Descriptive statistics ► Graphs

	Mean	P25	Median	P75	<i>N</i> (unw.)	<i>N</i> (wtd.)
Age	81.28	75.00	84.00	90.00	451,217	1,758,664
Female indicator	0.464	0.000	0.000	1.000	451,217	1,758,664
Married indicator	0.266	0.000	0.000	1.000	451,217	1,758,664
Other marital status indicator	0.212	0.000	0.000	0.000	451,217	1,758,664
Reliefs (£)	106,995.45	0.00	0.00	0.00	451,217	1,758,664
Tax due (£)	27,638.36	0.00	0.00	0.00	451,217	1,758,664
Total assets (£)	451,231.31	159,806.95	276,844.56	501,061.66	451,217	1,758,664
Tax rate	0.017	0.000	0.000	0.000	451,217	1,758,664
Zero tax liability indicator	0.869	1.000	1.000	1.000	451,217	1,758,664
Widowed indicator	0.522	0.000	1.000	1.000	451,217	1,758,664
Unexpected mortality risk (April)	0.461	0.408	0.473	0.514	451,217	1,758,664
Unexpected mortality risk (May)	0.264	0.205	0.300	0.335	451,217	1,758,664

Conditional unexpected death probability by region



Source: Authors' calculations using data from the ONS

Wealth gradient in excess mortality

- ▶ Potential selection issue if wealthier people are relatively more likely to die during COVID

Wealth gradient in excess mortality

- ▶ Potential selection issue if wealthier people are relatively more likely to die during COVID
- ▶ Relatively few studies on the wealth gradient in mortality during COVID, in the UK or other countries

Wealth gradient in excess mortality

- ▶ Potential selection issue if wealthier people are relatively more likely to die during COVID
- ▶ Relatively few studies on the wealth gradient in mortality during COVID, in the UK or other countries
- ▶ More on the income gradient in mortality:
 - ▶ Before the pandemic, clear negative income gradient in mortality rates (see, e.g., [Chetty et al., 2016](#))
 - ▶ During COVID: steeper negative income gradient ([Brandily et al., 2021](#); [Calderón-Larrañaga et al., 2020](#); [Chen et al., 2020](#))
 - ▶ Mediated by strong geographic location effects ([Decoster et al., 2021](#))

Wealth gradient in excess mortality

- ▶ Potential selection issue if wealthier people are relatively more likely to die during COVID
- ▶ Relatively few studies on the wealth gradient in mortality during COVID, in the UK or other countries
- ▶ More on the income gradient in mortality:
 - ▶ Before the pandemic, clear negative income gradient in mortality rates (see, e.g., [Chetty et al., 2016](#))
 - ▶ During COVID: steeper negative income gradient ([Brandily et al., 2021](#); [Calderón-Larrañaga et al., 2020](#); [Chen et al., 2020](#))
 - ▶ Mediated by strong geographic location effects ([Decoster et al., 2021](#))

↪ **We likely estimate a lower-bound of inheritance planning**